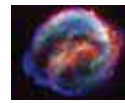




Sea turtles dying by the hundreds

As many as 300 to 400 dead sea turtles were found off the coast of El Salvador in Jiquilisco Bay late last month. <http://digit.in/RIPTurtles>



A star that won't die

A three-year-long supernova explosion challenges scientist's current understanding of star formation and death. <http://digit.in/Starnodie>

Man made earthquakes

As it turns out, it's our *fault* after all

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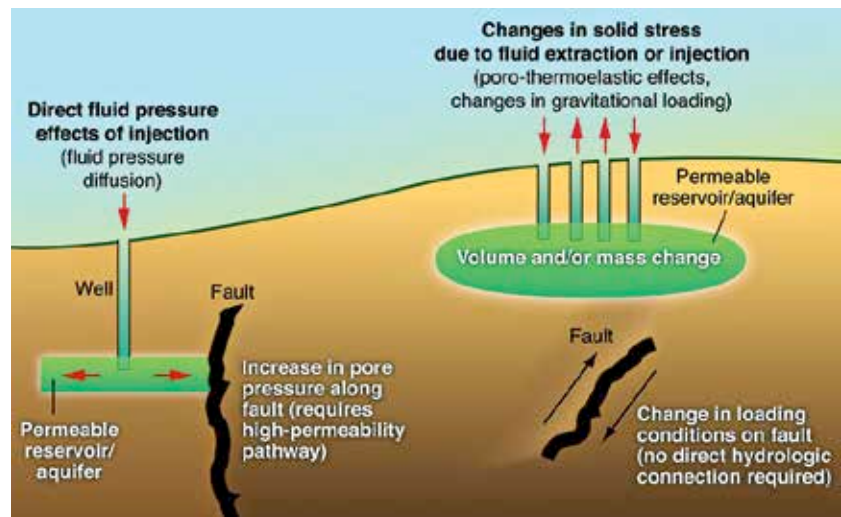
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arthquakes, have become a fairly common phenomenon in recent times. Even if that might not be the

case right here in India, United States has seen a significant rise in seismic activity. And no matter what we tell ourselves, there's an undeniable reason behind all of it – humans. But, wait a minute, human activity causing environmental change isn't exactly news to us, and neither is that helpful in determining their occurrence or what we can do about it – unless we know exactly what we're doing wrong. There are numerous studies into several different reasons that are behind induced seismicity, and its about time we knew what they're all about.

GROUNDWATER INJECTION IN OIL/GAS EXTRACTION

According to a 2015 study, between 1973 and 2008, the central part of United States saw an annual average of 24 earthquakes at or above the magnitude of 3.0 on the Richter scale. That number rose to an average of 193 earthquakes a year between 2009 and 2014. That is more than 700% growth in sheer numbers. The same study, by authors Justin Rubinstein of the U.S. Geological Survey (USGS) and Alireza Babaie Mahani of the Geological Survey of Canada, points out that the injection of fluids into the earth is one of main causes for the rise of



The effects that fluid injection and withdrawal can have on nearby faults can cause induced seismicity

seismic activity in the region. Which activities require putting liquids into the planet? Quite a lot of industrial processes actually. But a couple of the biggest culprits are oil and natural gas wells, which generate a significant amount of wastewater. Injecting waste water back into the ground increases subsurface pore pressure, causing faults to slip.

But we've been digging for oil and natural gas for quite a while now, so why the sudden rise? To state it simply, we're just doing it a lot more – thanks to all new extraction methods. One such method involves going after rock formations that haven't been previously identified as oil reservoirs, which generates as much as 15 to 50 times the water as compared to oil. As a result,

greater quantities of wastewater are being pumped into the earth than ever before. And this isn't as harmless as it was made out to be for a long time. On September 3, 2016, an earthquake with a magnitude of 5.8 occurred near Pawnee, Oklahoma, followed by nine aftershocks between magnitudes 2.6 and 3.6 within 3 1/2 hours.

ARTIFICIAL LAKES

Another significant factor contributing to man-made earthquakes are large and deep artificial water bodies – like the ones found in dams and reservoirs. In such reservoirs, the weight of the column of water is sufficient to alter the underlying fault or fracture condition by changing the overall stress on the same. It either happens due to an